7PAM2002 Data Science Project Module

Choosing a Project Form

Semester B 2025/2026

This form will be used by your supervisor to agree your project topic and dataset. Complete as much of the form as possible then submit the form into 'Assignments' on the Project Module Canvas site.

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Course: 2-year Advanced Research
Semester intake to the course: B

SECTION A	
What is your proposed project title or topic?	Lung Cancer Detection and Classification Using Deep Learning on the IQ-OTH/NCCD CT Scan Dataset
What is the Research Question for your project?	1.*Can deep learning models accurately detect and classify lung cancer (normal, benign, malignant) using CT scan images from the IQ-OTH/NCCD dataset?* 2. *Which deep learning architecture (CNN, transfer learning models, or advanced vision models) provides the highest accuracy and reliability for medical image classification?* 3. *How effectively can pre-processing and data augmentation techniques improve classification performance on limited medical imaging data?*
Dataset website address (or organisation and person the dataset is from).	Dataset: IQ-OTH/NCCD Lung Cancer CT Scan Dataset https://www.kaggle.com/datasets/hamdallak/the-iqothnccd-lung-cancer-dataset
Where was the data originally collected? (who, when, where)	**Organisation:** Iraq Oncology Teaching Hospital (IQ-OTH) **National Centre for Cancer Diseases (NCCD)** **Collected by:** Oncologists and radiologists at the above hospitals **When:** Over a 3-month period during Fall 2019 **Where:** Baghdad, Iraq (with cases from Baghdad, Wasit, Diyala, Salahuddin, and Babylon)
What type of data are you using? (e.g. image/ tabular/category/continuous etc)	**JPG images** of CT scan slices **Image-based medical data** (3 classes → normal, benign, malignant) Originally DICOM CT scans, converted into **JPG format**

	Each case contains multiple slices (80–200 images per patient)
SECTION B	
Can you attend all 6 supervision sessions on-campus? If not state reason.	**Yes, I can attend all the sessions on campus**
What was the subject of your BSc degree and any other master's degrees you have taken?	Civil Engineering (B Tech)
What is your career aims and/or the industry sector you would like to get into?	I would like to get into an organization that works on data domain, I want to become a Data Scientist.
A brief account of your programming/data science experience (including work or placement).	I am a civil engineering student with a strong interest in data science, and experience in Python-based data analysis, visualization, and basic machine learning. My engineering background has strengthened my ability to work with real-world datasets and solve practical problems using data.
Any work experience (including non-computing related).	I have one year of professional experience working as a Power BI Analyst, where I developed interactive dashboards and reports to support data-driven decision making. My role involved data cleaning, analysis, and visualization using Power BI and Excel, collaborating with stakeholders to understand business requirements
Do you have any hobbies, activities or interests.	My hobbies and interests include playing cricket and volleyball, which help me develop teamwork, discipline, and leadership skills.